0580_w24_ms_41_Q6

Question	Answer	Marks	Partial Marks	Guidance
6(a)	$\sqrt{420^2+830^2-2\times420\times830\times\cos106}$ oe	**M2**	or **M1** for $420^2+830^2-2\times420\times830\times\cos106$ oe **A1** for 1 057 474 ….	For M2 or M1 minimal acceptable expression is 865300 - 697200cos106 Note 1 057 474 …. following algebraic cosine rule with no substitutions shown scores 0
	1028.3...	**A1**		1028.3… after M1 scores M2A1
6(b)	99[.0] or 98.98 to 99.1[0…]	**4**	**B3** for 80.89 to 81.02 or **M2** for $\sin[ACB=]\frac{1150\sin62}{1028}$ oe or **M1** for $\frac{1028}{\sin62}=\frac{1150}{\sin ACB}$ oe	Allow more accurate value for 1028
6(c)	2477 cao nfww	**4**	**B3** for answer 2476.9… or **M2** for $\frac{1}{2}\times420\times830\times\sin106\times\frac{P}{10000}=41\,500$ oe or **M1** for $\frac{1}{2}\times420\times830\times\sin106$ oe	Nfww must be correct from their more accurate value if shown. Allow B3 for answer 2476 to 2478 Oe eg $\frac{41500}{\frac{1}{2}\times420\times830\times sin106\div10000}$ M1 implied by 168 000 or 167 500 or 167 550 or 167 547 to 167 548 Heron's formula area = $\sqrt{s(s-420)(s-830)(s-1028)}$ where $s=\frac{420+830+1028}{2}=1139$ Area = $\sqrt{1139\times719\times309\times111}$ = 167597.3071